# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A car starting from rest has acceleration 3 m/s squared. Its velocity after 4 s is (m/s): 1

**(a)** 7 m/s **(b)** 6 m/s **(c)** 1 m/s **(d)** 12 m/s

**2.** A force of 24 N acts on a body and produces an acceleration of 4 m/s squared. The mass is: 1

**(a)** 6 kg **(b)** 20 kg **(c)** 96 kg **(d)** 0.17 kg

**3.** A force of 20 N acts at right angles to the displacement of 5 m. The work done is: 1

**(a)** 100 J **(b)** 25 J **(c)** zero **(d)** 4 J

**4.** The frequency of a wave is 500 Hz and the speed is 340 m/s. The wavelength is: 1

**(a)** 0.0015 m **(b)** 0.68 m **(c)** 170000 m **(d)** 1.47 m

**5.** A body moves along a straight line with speed 4 m/s for 10 s and 6 m/s for the next 10 s. The average speed is: 1

**(a)** 4 m/s **(b)** 6 m/s **(c)** 10 m/s **(d)** 5 m/s

**6.** A person weighs 600 N on the Earth. On a planet with g one-third of the Earth's, the weight is: 1

**(a)** 1800 N **(b)** 300 N **(c)** 200 N **(d)** 600 N

**7.** A lamp of 60 W is used for 5 hours daily for 30 days. The energy consumed in kWh is: 1

**(a)** 1.8 **(b)** 300 **(c)** 9 **(d)** 90

**8.** Sound of two different tuning forks has frequencies 256 Hz and 512 Hz. The wavelength is: 1

**(a)** zero **(b)** equal

**(c)** greater for the 256 Hz fork **(d)** greater for the 512 Hz fork

**9.** A stone thrown vertically up returns to the thrower's hand after 6 s. The velocity of throw is (g = 10): 1

**(a)** 60 m/s **(b)** 10 m/s **(c)** 15 m/s **(d)** 30 m/s

**10.** A block of mass 4 kg rests on a rough table. The least force of 12 N makes it just move. Static friction is: 1

**(a)** 40 N **(b)** 4 N **(c)** 0 **(d)** 12 N

**11.** In a lever, the effort arm is 40 cm and the load arm is 10 cm. The MA is: 1

**(a)** 50 **(b)** 4 **(c)** 0.25 **(d)** 30

**12.** Two musical instruments produce notes of the same pitch and loudness but you can tell them apart. This is due to differing: 1

**(a)** frequency **(b)** quality or timbre
**(c)** amplitude **(d)** speed

**13.** A body starts from rest with acceleration 2 m/s squared. The distance covered in 10 s is: 1

**(a)** 200 m **(b)** 100 m **(c)** 20 m **(d)** 10 m

**14.** A body of mass 3 kg moving at 4 m/s is stopped in 2 s. The average force is: 1

**(a)** 3 N **(b)** 6 N **(c)** 24 N **(d)** 12 N

**15.** A skier of mass 60 kg slides down from rest through a vertical drop of 20 m. The speed at the bottom if friction is negligible (g = 10) is: 1

**(a)** 400 m/s **(b)** 40 m/s **(c)** 10 m/s **(d)** 20 m/s

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** A car moves with uniform velocity of 15 m/s for 20 s. Find the distance and draw the nature of the v-t graph. 2

**17.** A body of mass 5 kg is acted on by two forces: 20 N east and 8 N west. Find the acceleration. 2

**18.** A force of 40 N moves a body through 5 m in the direction of the force. Find the work done and the KE gained if the body starts from rest with mass 4 kg. 2

**19.** The frequency of a wave is 20 Hz and its wavelength 17 m. Find the speed and the time for 10 oscillations. 2

**20.** An object is dropped from 20 m. Find the time to hit the ground and the speed on impact (g = 10). 2

## SECTION — C (Short Answer Type) 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** A bus moving at 20 m/s speeds up uniformly to 30 m/s in 5 s. Find the acceleration, the distance and the average velocity. 3

**22.** A body of mass 10 kg is acted on by a force of 50 N for 4 s starting from rest. Find the acceleration, the final velocity and the distance covered. 3

**23.** A 2 kg body is thrown upward with 15 m/s. Find KE and PE at the start, at 5 m and at the top (g = 10). 3

**24.** A source produces 4000 vibrations in 10 s. Find the frequency, the time period and the wavelength in air (v = 340 m/s). 3

OR

A car A starts from rest with a = 2 m/s squared and car B moves at uniform 20 m/s. When will A overtake B if both start together from the same point? 3

**25.** A truck of mass 5000 kg moving at 10 m/s collides and stops in 0.5 s. Find the average force and compare with its weight. 3

OR

A lever has a load of 100 N at 20 cm from the fulcrum. Find the effort at 50 cm and at 1 m on the other side and compare MA. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** A car accelerates from rest at 2 m/s squared for 10 s, moves at constant velocity for 20 s and then retards uniformly to rest in 5 s. Find the maximum velocity, the retardation and the total distance. Draw the v-t graph in words. 5

OR

A car of mass 1000 kg moves on a level road at 30 m/s. The engine force is 2000 N and the resistive force is 1500 N. Find the acceleration. If the engine is switched off and the resistance remains, find the retardation, the time to stop and the stopping distance. 5

**27.** A body of mass 2 kg falls freely from 45 m. Find the PE and the KE at the start, after 1 s, 2 s and at the ground and show that the total energy remains constant (g = 10). 5

OR

A person standing between two parallel cliffs claps. He hears the first echo after 1 s and the second after 2 s. Find the distance between the cliffs and the distances from the person (v = 340 m/s). What would be the time if he is at the centre? 5

**END OF QUESTION PAPER** — **Total: 50 Marks**

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